Adnan Harun Doğan

Personal Information

Address:	Middle East Technical University, Office: A401, Ankara, Turkey
Email:	adnanharundogan@gmail.com
Personal Website:	adnanhd.github.io
Google Scholar:	scholar.google.com/citations?user=QGaRpqYAAAAJ
GitHub:	github.com/adnanhd
ORCID:	0000-0002-6802-6405

Employment

Feb 2023–Present — **Research Assistant**, Image Processing and Pattern Recognition Laboratory (ImageLab), METU
Supervisor: Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas — *Ankara, Turkey*

Jan 2025–Oct 2025 — **Intern**, Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich
Supervisor: Prof. Dr. Christian Holz — *Zürich, Switzerland*

Oct 2021–Feb 2022 — **Intern**, Center for Artificial Intelligence and Robotics (ROMER), METU
Supervisor: Assoc. Prof. Dr. Erol Sahin, Assoc. Prof. Dr. Bugra Koku — *Ankara, Turkey*

Jun 2021–Sep 2021 — **Intern**, Cyber-Physical Systems Laboratory (AICPS), University of California, Irvine
Supervisor: Prof. Dr. Mohammad Abdullah Al-Faruque — *Irvine, California, USA*

Education

2022–2025 — **Master of Science in Computer Engineering**, Middle East Technical University (METU)
Advisor: Prof. Dr. Sinan Kalkan, Co-advisor: Assoc. Prof. Dr. Emre Akbas — *Ankara, Turkey*
Thesis: Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation

2018–2022 — **Bachelor of Science in Computer Engineering**, Middle East Technical University (METU)
Senior Project: ReQuest: Automated Question-Answer Generation from Text — *Ankara, Turkey*

Research Interests

My research focuses on differentiable optimization: embedding combinatorial and constrained solvers, such as ranking, matching, and assignment, as layers of neural networks. Because the solutions of such solvers change discontinuously with their inputs, standard backpropagation provides little or no gradient signal. I am interested in principled surrogate gradients that respect the structure of the underlying problem, and in understanding how this choice influences what the network learns. My main application area is computer vision, in particular object detection, where ranking-based losses and optimal-transport matching let detectors be trained directly against the criteria by which they are evaluated.

Keywords: differentiable optimization, combinatorial optimization, optimal transport, deep learning, object detection

Publications

Peer-reviewed publications

10. **Dogan, A. H.**, Demirel, B. U., & Holz, C. (2026). Frequency-weighted neural Kalman filters. ***International Conference on Robotics and Automation (IEEE ICRA)***. arXiv:2606.02251 (PDF, Code)
9. **Dogan, A. H.**, Deniz, M., Alemdar, H., & Baran, O. U. (2026). CoFINN: Conservation flux informed neural networks for physics problems governed by conservation laws. *Scientific Reports*. arXiv:2607.06587
8. Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus, M., & Holz, C. (2025). Beyond subjectivity: Continuous cybersickness detection using EEG-based multitaper spectrum estimation. ***IEEE Transactions on Visualization and Computer Graphics (IEEE TVCG), 31***(5), 2525-2534. doi.org/10.1109/TVCG.2025.3549132 (arXiv, Code)
7. Yavuz, F., Cam, B. C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., & Akbas, E. (2024). Bucketed ranking-based losses for efficient training of object detectors. ***European Conference on Computer Vision (ECCV)***, Lecture Notes in Computer Science, vol 15117, pp. 93-109. doi.org/10.1007/978-3-031-73202-7_6 (arXiv, PDF, Code)
6. Amin, M. A., **Dogan, A. H.**, Kuru, E. S., Sever, Y., & Angin, P. (2024). Misuse detection and response for orchestrated microservices based software. *International Conference on Advanced Information Networking and Applications (AINA)*, Lecture Notes on Data Engineering and Communications Technologies, vol 204, pp. 217-226. doi.org/10.1007/978-3-031-57942-4_22 (Data)
5. Karagoz, P., Cekinel, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, T. S., & Tuncel, B. M. (2023). Enhancing underground built heritage analysis with text mining: A case study on Cappadocia. In M. Akkar Ercan & K. Aydiner (Eds.), *Valorising Underground Built Heritage in Cappadocia* (pp. 61-92). CNR Edizioni. ISBN 978-88-8080-605-9. (PDF)
4. Sever, Y., & **Dogan, A. H.** (2023). A Kubernetes dataset for misuse detection. *ITU Journal on Future and Evolving Technologies, 4*(2), 383-388. doi.org/10.52953/fplr8631 (PDF, Code)
3. Sever, Y., Ekinci, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., & Angin, P. (2022). An empirical analysis of IDS approaches in container security. *International Workshop on Secure and Reliable Microservices and Containers (SRMC)*, pp. 18-26. doi.org/10.1109/SRMC57347.2022.00007 Best Paper Award
2. Demirel, B. U., **Dogan, A. H.**, & Al-Faruque, M. A. (2021). Two might do: A beat-by-beat classification of cardiac abnormalities using deep learning and domain-specific features. *Computing in Cardiology (CinC)*, pp. 1-4. doi.org/10.23919/cinc53138.2021.9662935 (PDF, Code)
1. Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., & Karagoz, P. (2021). Explainability in irony detection. *Big Data Analytics and Knowledge Discovery (DaWaK)*, Lecture Notes in Computer Science, vol 12925, pp. 152-157. doi.org/10.1007/978-3-030-86534-4_14

Manuscripts under review

2. **Dogan, A. H.**, Arasli, D., Oksuz, K., Akbas, E., & Kalkan, S. (2027). *Variational surrogate Jacobians: A framework for constrained gradient propagation.* Under Review[†].
1. **Dogan, A. H.**, Celtik, S. O., Baran, O. U., & Alemdar, H. (2026). *Prediction-driven adaptive mesh refinement with graph neural networks for transonic airfoil flows.* Under Review at Journal of Aerospace Science and Technology.

Theses

1. **Dogan, A. H.** (2025). *Incorporating piecewise linear functions with constant regions in backpropagation.* M.Sc. Thesis, Middle East Technical University. (Thesis)

[†] Venue withheld while under double-blind review.

Research Experience

Sep 2021–Jun 2022

High Performance Intrusion Detection and Prevention for Software-Defined Container Networks in the Cloud, METU

Undergraduate Research Project

TUBITAK 3501 funded research project. Designed and implemented high-performance IDS pipeline optimized for containerized cloud infrastructures. Collaborated on Kubernetes misuse detection research, contributing to novel datasets and detection methodologies.

Advisor: Assoc. Prof. Dr. Pelin Angin *Ankara, Turkey*

Jan 2020–Jun 2022

Image Processing and Pattern Recognition Laboratory (ImageLab), METU

Undergraduate Research Project

CNNFOIL: Approximating HLLC Riemann solver for flow prediction around airfoils via encoder-decoder neural networks. Engineered state-of-the-art neural network models achieving 4.4% performance improvement in predicting critical flow field parameters around airfoil geometries.

Advisor: Assist. Prof. Dr. Hande Alemdar, Assist. Prof. Dr. Ozgur Ugras Baran *Ankara, Turkey*

Oct 2020–Oct 2021

Textual Event Graph HUB (TEGHub), METU

Undergraduate Research Project

Textual analysis for irony detection and research impact prediction. Implemented end-to-end neural pipelines integrating GCN and LSTM networks for analyzing citation patterns and textual content on Twitter datasets.

Advisor: Prof. Dr. Pinar Karagoz *Ankara, Turkey*

Feb 2020–Jun 2021

School of Medicine, Koç University

Undergraduate Research Project

Healthcare policy simulation for evidence-based decision making. Utilized simulation models focusing on mechatronics applications in healthcare. Developed GCN-based citation graph embeddings for healthcare policy papers.

Advisor: Assist. Prof. Dr. Ozge Karanfil *Istanbul, Turkey*

Teaching

Teaching Assistant, Computer Engineering Department, METU Feb 2022–Present

8. CENG 280 - Automata Theory (Spring 2024, Spring 2025, Spring 2026)
7. CENG 242 - Programming Language Concepts (Spring 2022, Spring 2023, Spring 2025, Spring 2026)
6. CENG 223 - Discrete Computational Structures (Fall 2023, Fall 2024, Fall 2025)
5. CENG 424 - Logic for Computer Science (Fall 2023, Fall 2025)
4. CENG 460 - Introduction to Robotics for Computer Science (Fall 2024)
3. CENG 213 - Data Structures (Spring 2024)
2. CENG 382 - Analysis of Dynamical Systems (Spring 2023)
1. CENG 334 - Introduction to Operating Systems (Spring 2022)

Undergraduate Teaching Assistant, Computer Engineering Department, METU Spring 2019–Fall 2021

3. CENG 240 - Programming with Python for Engineers (Fall 2021)
2. CENG 111 - Introduction to Computer Engineering Concepts (Fall 2020)
1. CENG 230 - Introduction to C Programming (Spring 2019)

Awards and Fellowships

2022–2023	TUBITAK 2247-C STAR'23 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2022	Graduated in the first quantile of the class by Middle East Technical University (METU)
2021–2022	TUBITAK 2247-C STAR'22 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2020–2021	TUBITAK 2247-C STAR'21 Scholarship by The Scientific and Technological Research Institution of Turkey (TUBITAK)
2017	University Entrance Exam: ranked 1466th of 2.27M candidates by OSYM, Turkey

Computer Skills

Languages & Tools:	Python/PyTorch, C/C++, ROS2, Rust
DevOps & HPC:	GitHub, Docker, Slurm (Workload Manager), Apptainer

Languages

Turkish:	Native
English:	Fluent (TOEFL iBT 109/120, 2022)